

AdaWGAN: Data Augmentation for Few-Shot HD-sEMG Gesture Recognition Using Single-Trial Data

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Abstract—Despite recent advances in human-computer interaction (HCI) based on physiological signals like surface electromyography (sEMG), the scarcity of high-quality labeled data remains a major barrier for human intention recognition and practical HCI deployment. Existing sEMG augmentation methods often use temporal or frequency-domain representations from sparse electrode recordings, while the spatial muscle activation patterns in high-density sEMG (HD-sEMG) remain insufficiently exploited. To address this challenge, we propose a dual-branch Adaptive Weight Wasserstein Generative Adversarial Network (AdaWGAN) for physiologically consistent HD-sEMG augmentation. AdaWGAN jointly learns RMS-based spatiotemporal activation maps and frequency-band features, enabling the generation of high-fidelity, category-aligned HD-sEMG samples from single-trial training data for gesture recognition. Furthermore, an adaptive weighting mechanism is incorporated into the generator objective to dynamically balance adversarial and classification losses, thereby enhancing the semantic consistency between real and generated samples. Compared with state-of-the-art generative methods, the synthesized samples of AdaWGAN exhibited high fidelity to real data, achieving Pearson correlation coefficients exceeding 0.95 for most gesture classes. Evaluations on two benchmark HD-sEMG datasets demonstrated that AdaWGAN achieved the best gesture classification accuracies of $88.2 \pm 6.3\%$ and $94.3 \pm 4.4\%$, representing an improvement of approximately 5% over the baseline and outperforming other augmentation strategies. Furthermore, attribution visualization analysis indicated that AdaWGAN captures physiologically meaningful features consistent with genuine muscle activation patterns. This

work provides an effective solution for sEMG data augmentation and contributes to the development of interpretable and physiologically plausible generative models for biomedical HCI.

Index Terms— HD-sEMG, data augmentation, generative adversarial networks, adaptive weighting, physiological interpretability.

I. INTRODUCTION

RECENT advances in artificial intelligence and embodied intelligence have enabled more precise control of Human-Computer Interaction (HCI) systems [1]. In this context, surface electromyography (sEMG) interfaces have emerged as a vital solution, offering intuitive, high-fidelity motor control while ensuring long-term wearing comfort [2]. EMG-based gesture recognition relies on large-scale, high-quality sEMG data. However, the collection of sEMG signals involves repetitive movements, which can be tedious for subjects. This repetition not only causes participant fatigue but also disrupts electrode-skin contact, potentially leading to electrode offset and distorted signals [3]. Consequently, acquiring a large amount of high-quality sEMG samples remains a challenging task [4]. Additionally, privacy regulations related to subjects also contribute to the limited availability of sEMG datasets [5]. It is thus critical to develop effective and reliable data augmentation methods that can generate large-scale, high-quality sEMG data for precise human-computer interaction.

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Traditional data augmentation methods include rotation, shifting, cropping, and scaling [6]. These approaches apply geometric transformations to create more data. While they are simple, efficient, and can enhance model generalization and robustness [7], they are typically designed for images and are not suitable for sEMG, a type of time series data. Common data augmentation methods for time-series data, including flipping, window warping, and adding noise [8], [9], have limited applicability to sEMG. These methods can disrupt their inherent temporal dependencies and muscle activation profiles, and may even introduce physiologically implausible artifacts [10].

By learning the underlying distribution and correlations in training data, generative adversarial networks (GANs) can generate complex EMG data that is physiologically plausible. Lin et al. [11] proposed the RoHDE framework to generate noisy high-density surface EMG (HD-sEMG) signals to enhance model robustness. By synthesizing data that simulates interference such as contact artifacts and electrode leakage, their framework enabled pattern recognition systems to improve robustness in noisy environments. Zhao et al. [12] introduced E-TRGAN, a transformer-based GAN network, for the efficient reconstruction of impaired HD-sEMG signals while preserving their spatiotemporal structure. These works demonstrate the feasibility of using GANs to generate sEMG data. However, neither was used for the task of small-sample data augmentation. For few-shot datasets enhancement, recent work by Nasrallah et al. [5] showed that transformer-based CGANs can generate sEMG data to effectively augment small or imbalanced datasets, improving classification accuracy. Zhang and Ma [13] proposed a time domain sEMG feature generation method based on an energy-based generative adversarial network (EBGAN) that effectively captured the distribution of real datasets to generate high-quality synthetic samples. Both methods, however, are constrained by inadequate input representations for focusing solely on either temporal or frequency domain. This lack of rich input data ultimately limits their performance in replicating complex patterns and generalizing to new tasks. Furthermore, the generation of sEMG signals should align with the physiological structures of the human body, given its direct relationship to neuromuscular activation patterns. Current methods, however, fails to provide a physiological rationale for the generated signals, which limits their credibility and clinical applicability.

In this paper, we propose an Adaptive Weight Wasserstein GAN (AdaWGAN) to synthesize high-fidelity and category-aligned HD-sEMG samples using single-trial training data, aiming to improve gesture recognition performance in few-shot scenarios. To ensure semantic consistency between real and generated samples, we incorporate an adaptive weighting mechanism into the generator's objective function, which dynamically balances adversarial and classification losses. Simultaneously, to capture comprehensive signal information, we employ a dual-branch architecture in the generator and discriminator to encode input time-domain and frequency-

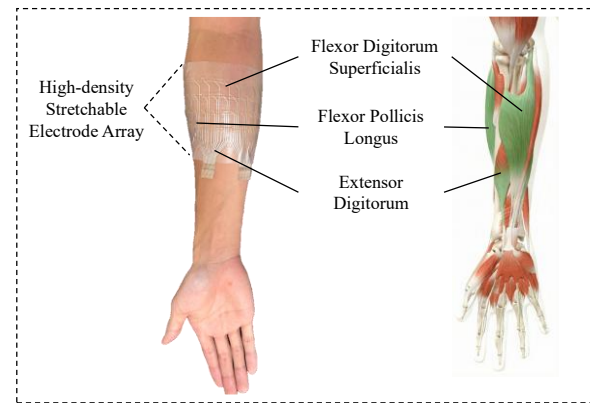


Fig. 1. The attached electrodes correspond to the positions of the muscles.

domain features. Consequently, under single-trial training conditions, the proposed method improved gesture classification accuracies on two benchmark datasets from $83.6 \pm 7.9\%$ and $90.8 \pm 7.4\%$ to $88.2 \pm 6.3\%$ and $94.3 \pm 4.4\%$, respectively. Furthermore, to address the challenge of interpretability, we verify the physiological reliability of our model using convolutional kernel activation visualization and Integrated Gradients, confirming that the synthetic samples preserve authentic muscle activation patterns.

II. METHODOLOGY

A. Datasets and Preprocessing

1) **SIAT Dataset:** We employed a self-developed high-density, flexible, and stretchable electrode array [14], [15] to record sEMG signals from subjects performing hand gestures. As shown in Fig. 1, the array consists of a 4×16 grid of channels, which are spaced 1 cm apart and attached to the subjects' right forearms over the muscles near the elbow, covering the finger flexors and finger extensors. EMG signals were recorded at a rate of 2000 Hz, using a high-performance acquisition system (NES-64B01, SIAT, China). During acquisition, the EMG signal was amplified and filtered with a notch filter at 50Hz and a bandpass filter between 20 and 500Hz, as EMG signal energy predominantly lies in this range [16].

The dataset comprised seven able-bodied subjects aged 23 to 30. During the experiment, each subject was instructed to perform six specific hand gestures following visual instructions on the computer screen. These gestures included hand close (HC), hand open (HO), wrist extension (WE), and wrist flexion (WF), wrist pronation (WP), wrist supination (WS). During the experiment, each movement was held for 4 seconds and repeated six times. To avoid potential muscle fatigue, a 4-second rest was incorporated between two consecutive repeats.

The experimental protocol was approved by the Ethics Committee of the Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, in accordance with the Declaration of Helsinki, and the approval number was SIAT-IRB-240415-H0855.

2) **The Public HD-sEMG-65 Dataset:** The public high-density surface electromyography (HD-sEMG) dataset by Malešević et al. [17] was employed in this study. We utilized

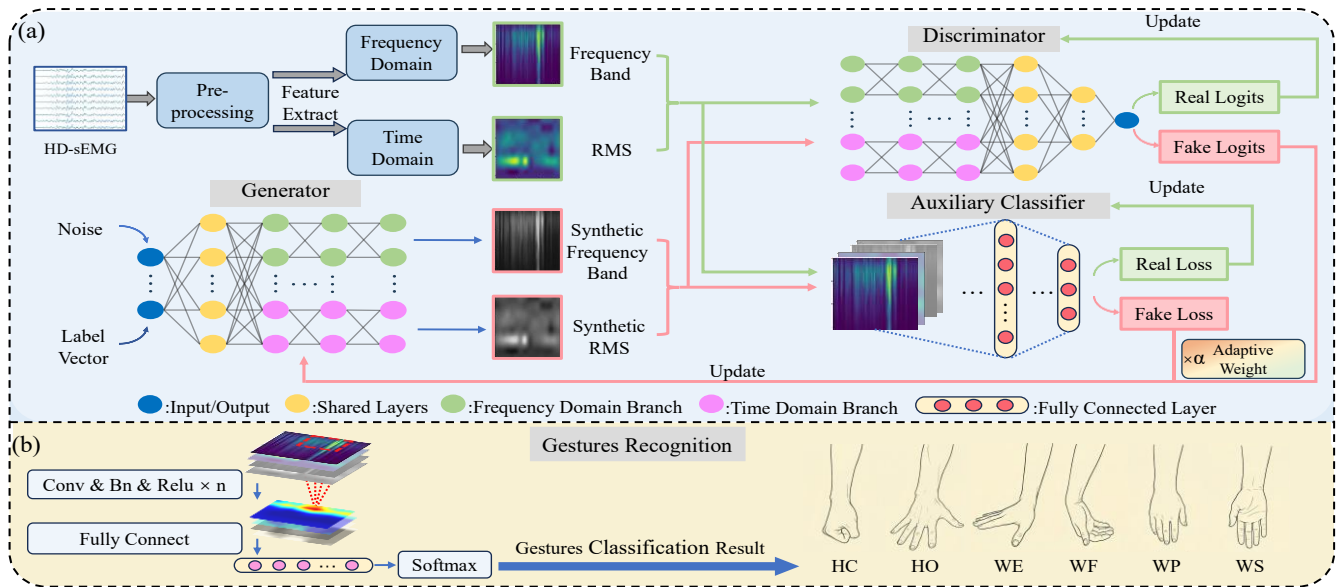


Fig. 2. (a) AdaWGAN framework and training procedure. The real training data, consisting of feature bands and RMS features, is used to update the weights of the discriminator and auxiliary classifier. The generator utilizes noise and label guidance to synthesize samples, which are then evaluated by the discriminator and classifier to optimize the generator's weights. (b) Using the trained generator to enhance datasets for gestures recognition.

data from seven subjects. The EMG signals were acquired using two 8×8 electrode arrays (totaling 128 channels) placed on the dorsal and palmar sides of the forearm, with a sampling rate of 2048 Hz. The dataset was band-pass filtered with a hardware high-pass filter at 10 Hz and a low-pass filter at 900 Hz during recordings. Seven single-degree-of-freedom movements were selected as experimental samples, including little finger bend (LB), ring finger bend (RB), middle finger bend (MB), index finger bend (IB), thumb down (TD), wrist bend (WB), and wrist stretch (WS). Each posture was repeated for five times.

B. The Proposed Adaptive Weight Wasserstein GAN

1) **Model Architecture:** We propose a novel generative network called AdaWGAN, designed to generate synthetic samples with high fidelity to real data. This design is based on a Wasserstein GAN with gradient penalty (WGAN-GP) [18], owing to its stable training behavior and demonstrated ability to produce high-quality samples. The overall architecture of AdaWGAN is shown in Fig. 2(a).

The AdaWGAN consists of three major components: the Generator, the Discriminator and the Classifier. The Generator is responsible for synthesizing highly realistic samples by learning from the input data. The Discriminator aims to distinguish between real samples from the training distribution and synthetic samples produced by the Generator. The Classifier serves as an auxiliary component that provides class guidance to the Generator. By classifying both real and synthetic samples according to their gesture labels, it ensures that the generated samples are semantically consistent with their target classes.

For training data preparation, we employ two feature extraction methods, as depicted in Fig. 2(a). To capture frequency information, we utilize the frequency-band (FB) characteristics of EMG signals. These features can effectively indicate muscle exertion levels [19]. In this process, we divide

the band-pass filtered EMG signals into 16 equal frequency bands across all channels to extract precise frequency details. For spatiotemporal information, we adopt the root mean square (RMS) features and arrange them spatially according to the matrix of electrodes. Using a sliding window of 200ms with a 50ms step size, the RMS value is computed as follows:

$$F_{RMS} = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2} \quad (1)$$

where F_{RMS} denotes the extracted RMS feature, N is the total number of samples within a window, and x_i represents the signal amplitude of the i -th sample.

Both feature sets are resized to 64×64 via linear interpolation to serve as input data. Leveraging these features ensures the model captures sufficient information within the EMG signals, and generates samples close to the real-sample distribution.

2) **Generator with Adaptive Weights:** The generator employs a dual-branch architecture to extract and utilize both spatial and temporal characteristics embedded in HD-sEMG data (as detailed in Table I), thereby facilitating the generation of high-fidelity samples. For the input, in addition to the Gaussian noise z , we incorporate a K -class label guidance vector. This vector provides categorical information, ensuring the generated samples are aligned with the target gestures.

To address the potential training imbalance between adversarial generation and category classification within the multi-task learning framework, we introduce an adaptive weighting mechanism inspired by GradNorm [20]. During training, the gradients of the adversarial loss are often more volatile and larger in magnitude than those of the classification loss, which can cause the generator to neglect the class condition. Unlike static weights, this mechanism dynamically adjusts the weight α by computing the L_2 norms of the

TABLE I
AdaWGAN'S MODEL PARAMETERS

Net	Number	Input Size	Layer	Output Padding	Bn	Dropout	Activation Function	Output Size
G	Embedding	K	Embedding	—	—	—	—	64
	Shared Layer	64+64	Linear (128, 8192)	—	No	No	ReLU	8192
	1	[128, 8, 8]	Deconv2d (128, 64, 3, 2, 1)	1	Yes	0.5	ReLU	[64, 16, 16]
	2	[64, 16, 16]	Deconv2d (64, 32, 3, 2, 1)	1	Yes	0.5	ReLU	[32, 32, 32]
	3	[32, 32, 32]	Deconv2d (32, 1, 3, 2, 1)	1	Yes	0.5	Tanh	[1, 64, 64]
D	1	[1, 64, 64]	Conv2d (1, 32, 3, 2, 1)	—	Yes	0.5	LeakyReLU	[32, 32, 32]
	2	[32, 32, 32]	Conv2d (32, 64, 3, 2, 1)	—	Yes	0.5	LeakyReLU	[64, 16, 16]
	3	[64, 16, 16]	Conv2d (64, 128, 3, 2, 1)	—	Yes	No	LeakyReLU	[128, 8, 8]
	Shared Layer_1	[256, 8, 8]	Conv2d (256, 256, 3, 2, 1)	—	Yes	0.5	LeakyReLU	[256, 4, 4]
	Shared Layer_2	[256, 4, 4]	Conv2d (256, 512, 3, 2, 1)	—	Yes	0.5	LeakyReLU	[512, 4, 4]
	Shared Layer_3	[512, 4, 4]	Flatten&Linear (8192, 1)	—	No	No	—	1
C	1	[2, 64, 64]	Flatten&Linear (8192, 2048)	—	Yes	0.6	ReLU	2048
	2	2048	Linear (2048, 512)	—	Yes	0.6	ReLU	512
	3	512	Linear (512, K)	—	No	No	Softmax	K

We denote the generator, discriminator and classifier as G, D and C, respectively. K represents for class number.

gradients for both loss functions with respect to the generator's parameters. This ensures that the magnitude of the classification-task gradients is scaled up to the same level as those of the adversarial task. We incorporated a numerical stability constant ϵ and a clipping threshold δ_{max} to prevent gradient explosion during early training. The final generator optimization objective and the calculation of the adaptive weight are as follows:

$$L_G = -E_{\tilde{x} \sim P_g}[D(\tilde{x})] - \alpha \cdot E[\log p(C = c|X_{fake})] \quad (2)$$

$$\alpha = clip\left(\frac{\|\nabla_{\theta_G} L_D\|_2}{\|\nabla_{\theta_G} L_C\|_2 + \epsilon}, \delta_{max}\right) \quad (3)$$

We define $\tilde{x}$ as a generated sample from the generator distribution P_g . C denotes the output of the classifier and c represents the ground truth. $D(\cdot)$ represents the discriminator. In formula (3), ∇_{θ_G} represents the gradient operator with respect to the generator's parameters θ_G . L_D and L_C are losses of the discriminator and classifier, respectively. $clip(\cdot)$ is a truncation function that restricts the value within the range $[0, \delta_{max}]$, and we set $\delta_{max} = 100$ in our experiments.

3) Discriminator and Classifier Network: Considering the two feature domains as inputs, we implemented a dedicated design with separate processing streams for each feature in the discriminator, as detailed in Table I. In the initial layers, separate branches are employed to identify and encode key patterns from the frequency and time domains, respectively. Subsequently, we utilize shared layers to fuse these branches and extract comprehensive representations. This design enables the discriminator to perform a more detailed characterization of the input samples. Regarding the objective function, we define x as a real sample and P_r as the real data distribution. A gradient penalty is added to the discriminator loss function to prevent gradient explosion. Let $\hat{x}$ be a random interpolation point between P_r and P_g , and $P_{\hat{x}}$ denote the distribution corresponding to $\hat{x}$. ∇ represents the gradient operator, and λ is

the penalty coefficient. The discriminator loss L_D is given by the following formula:

$$L_D = E_{\tilde{x} \sim P_g}[D(\tilde{x})] - E_{x \sim P_r}[D(x)] + \lambda E_{\hat{x} \sim P_{\hat{x}}}[(\|\nabla_{\hat{x}} D(\hat{x})\|_2 - 1)^2] \quad (4)$$

Unlike methods that append a classification branch to the output of the discriminator [21], [22], we employ an independent classifier to predict the category information of input samples. This strategy avoids conflicting objectives between the discriminative and classification tasks, which can lead to poor generation [23], [24]. To ensure the reliability of the predictions, the classifier is trained exclusively on real data and their corresponding labels. The classifier loss L_C based on cross-entropy is defined as follows:

$$L_C = -E_{x \sim P_r}[\log p(C = c|X_{real})] \quad (5)$$

4) Training Setup: All models were implemented using the PyTorch framework, and the training process was accelerated by an NVIDIA TITAN V GPU. The generative network was trained for 1,000 epochs, at which point the training loss had fully converged in the pre-experiments. The discriminator, generator, and classifier were all optimized using the Adam optimizer with $\beta^1 = 0.9$, $\beta^2 = 0.999$, a learning rate of $5e-4$, and a batch size of 64. To simulate a few-shot dataset scenario, we employed a leave-one-trial-out strategy per subject for both datasets. In this setup, the generator was trained on all samples of a single trial, with the remaining trials used as the test set. For each trial, we trained a corresponding generator, the all generators were then used to augment the simulated few-show HD-sEMG dataset.

C. Evaluation for Different Augmentation Methods

1) Evaluation Protocol: We investigated the impact of data scarcity and the efficacy of varying augmentation ratios for gesture classification by comparing different training data configurations. Specifically, we established three conditions:

Few-shot Condition: The evaluation classifier was trained using real data from a single trial only without any augmentation, referred to as "Single Trial".

Standard Condition: The evaluation classifier was trained using real data from four trials without any augmentation, denoted as “Four Trials”.

Augmented Condition: The single-trial real data was combined with synthetic data. To analyze the effect of dataset size, we generated synthetic samples equivalent to 50%, 100%, and 200% of the real data volume (one trial), denoted as “1:1/2”, “1:1”, and “1:2”, respectively.

2) Comparative Methods: We selected the AC-WGAN-GP framework by Sun et al. [25] as an ablation baseline to evaluate the specific contribution of the adaptive weighting mechanism. Its architecture mirrors that of AdaWGAN but excludes the adaptive weight. Furthermore, to verify the necessity of generative modeling for gesture recognition tasks, we benchmarked our approach against a standard non-generative method [26] that utilizes rotation and offset operations, referred to here as Geometric Augmentation.

3) Performance Metrics and Statistical Analysis: To assess the quality of the synthetic data and quantify its impact on the gesture classification task, we employed a six-layer Convolutional Neural Network (CNN) to evaluate the performance improvement achieved by using synthetic data. We used the classification accuracy (CA) (formula (6)) and the confusion matrix to evaluate the gesture classification performance. In this study, we adopted a K -fold cross-validation scheme to evaluate the model performance, where K represents the number of trials for each subject. In each fold, the classifier was trained using the i -th trial, while the remaining trials served as the test set. The classification result was the average of K trials.

$$CA = \frac{\text{correctly classified samples}}{\text{the whole test samples}} \times 100\% \quad (6)$$

Finally, statistical significance was assessed using paired t -tests on classification accuracy, with the significance level set to $p = 0.05$.

D. Fidelity Evaluation and Model Interpretation

1) Visualization and Correlation Analysis: To qualitatively assess the fidelity of the generated samples, we visualized the Frequency-Band (FB) and Root Mean Square (RMS) feature maps, as detailed in Section II-B. The Proposed Adaptive Weight Wasserstein GAN. All feature maps are normalized to the range of $[0,1]$.

We employed t -distributed Stochastic Neighbor Embedding (t -SNE) to visualize the distribution of both real and synthetic data. The projections are plotted on a unified scale, using distinct marker shapes and colors to differentiate between data sources and gesture categories. Additionally, to compare the impact of different generation methods and real-to-synthetic data ratios, we present visualizations across four specific ratios for both the proposed AdaWGAN and the comparative AC-WGAN-GP.

To evaluate the structural consistency between real and generated features, we computed the Pearson correlation coefficient based on the mean intensities of the FB and RMS features. Specifically, we averaged the feature intensities

separately for the real and generated samples across each subject. The correlation was then calculated for RMS features, FB features, and the concatenated RMS-FB features, denoted as RMS r , FB r , and Total r , respectively:

$$r = \frac{Cov(R, G)}{\sigma_R \sigma_G} \quad (7)$$

where $Cov(R, G)$ denotes the covariance of the mean intensities between the real and generated samples, and σ_R and σ_G represent their respective standard deviations.

To assess the distributional change after augmentation, we calculated the intra-class feature distance:

$$D = \frac{1}{N} \sum \|z_i - z_j\|_2 \quad (8)$$

where z_i and z_j are feature vectors from the same gesture class, and N is the number of sample pairs.

2) Model Interpretation: Attribution visualization analyses were applied to the discriminator to verify the design reasonableness of the proposed AdaWGAN network. In this framework, the discriminator not only assesses the authenticity of input samples but also provides the adversarial loss that directly guide the generator's gradient updates. Therefore, its internal feature responses can effectively reflect the model's learning mechanism toward the input data. Considering that shallow convolutional layers typically learn low-level features such as edges and textures, we first averaged the activation responses across all kernels of the first convolutional layer in both the RMS and frequency-band branches to obtain branch-specific activation maps (AM). To further examine deeper spatial-frequency learning, we extracted activation maps from the first three convolutional layers of the frequency-band branch. Shared layers were excluded because they fuse RMS and frequency-band features and no longer represent frequency-specific responses independently. Additionally, we employed the Integrated Gradients (IG) method [27] to perform a global attribution analysis on the discriminator. IG quantifies pixel-wise contributions to the model's decision by integrating gradients from a baseline to the input, revealing the spatial distribution of critical discriminative features. The formula is provided below:

$$IG_i(x) = (x_i - x'_i) \times \int_{\alpha=0}^1 \frac{\partial F(x' + \alpha(x - x'))}{\partial x_i} d\alpha \quad (8)$$

Here, $IG_i(x)$ denotes the attribution value of the i -th pixel, calculated by integrating the gradient $\frac{\partial F}{\partial x_i}$ along the path $x' + \alpha(x - x')$. This path is a linear interpolation from the all-zero baseline x' to the input x ($\alpha \in [0,1]$), where the term $(x_i - x'_i)$ scales the result back to the original input space.

III. RESULTS

A. Gesture Classification Performance and Augmentation Analysis

Table II and III present the gesture recognition results on the two datasets under various data augmentation settings. The terms “Single Trial” and “Four Trials” correspond to the Few-Shot Condition and Standard Condition, respectively, as

TABLE III
GESTURE CLASSIFICATION PERFORMANCE OF DIFFERENT AUGMENTATION METHODS UNDER VARIOUS REAL-TO-SYNTHETIC RATIOS ON THE SIAT DATASET

Subject	Single Trial	Four Trials	Geometric Augmentation (Real: Synthetic)			AC-WGAN-GP (Real: Synthetic)			AdaWGAN (Real: Synthetic)		
			1:1/2	1:1	1:2	1:1/2	1:1	1:2	1:1/2	1:1	1:2
1	95.6±0.2	96.7±0.5	95.8±0.1	95.6±0.2	95.3±0.2	95.4±0.2	95.7±0.1	95.9±0.1	97.9±0.5	97.8±0.7	97.8±0.4
2	85.1±1.5	91.6±1.8	85.7±1.3	85.1±1.3	84.5±1.4	86.7±1.2	86.7±1.2	86.2±1.3	90.3±1.3	90.5±1.6	90.2±1.3
3	80.6±3.1	86.4±2.8	82.9±2.9	82.2±3.0	81.9±3.1	82.8±3.7	82.8±2.7	82.5±2.7	83.0±3.3	82.2±2.8	82.4±3.4
4	79.1±3.9	88.6±1.8	76.5±3.9	77.0±3.8	77.2±4.0	76.0±3.4	79.5±4.0	80.9±3.7	86.1±3.5	84.9±3.7	85.5±4.1
5	87.5±2.4	93.8±1.2	88.1±2.3	88.4±2.9	88.0±2.3	88.5±2.4	89.0±2.3	89.0±2.3	91.5±2.1	91.4±2.0	91.0±1.9
6	70.4±6.6	86.8±2.0	70.3±5.7	70.3±5.6	70.9±6.5	71.4±7.1	69.4±6.9	73.8±6.2	78.4±7.1	78.3±6.4	77.7±7.1
7	87.3±2.3	94.0±1.3	88.3±2.3	88.0±2.2	88.1±2.3	87.8±1.9	88.9±1.8	88.9±1.2	89.9±1.7	89.3±2.1	89.3±2.0
Avg	83.6±7.9	91.1±4.0	84.0±8.4	83.8±8.3	83.8±8.0	84.1±8.1	84.6±8.5	85.3±7.1	88.2±6.3	87.8±6.5	87.7±6.5
Sig			--	--	--	--	--	*	*	*	*

“Single Trial” and “Four Trials” indicate the classifiers were trained using data from single trial and four trials, respectively. “Real: Synthetic” means the ratio between the real data and synthetic data. “Sig” represents the significances which were conducted between the “Single Trial” and different augmentation methods (* $p < 0.05$, “--”: not significant).

TABLE II
GESTURE CLASSIFICATION PERFORMANCE OF DIFFERENT AUGMENTATION METHODS UNDER VARIOUS REAL-TO-SYNTHETIC RATIOS ON THE PUBLIC HD-sEMG-65 DATASET

Subject	Single Trial	Four Trials	Geometric Augmentation (Real: Synthetic)			AC-WGAN-GP (Real: Synthetic)			AdaWGAN (Real: Synthetic)		
			1:1/2	1:1	1:2	1:1/2	1:1	1:2	1:1/2	1:1	1:2
1	97.9±0.5	97.6±1.4	97.6±0.8	97.5±0.7	97.4±0.8	96.6±0.3	96.5±0.5	97.6±0.8	98.3±1.0	98.5±1.0	98.4±1.1
2	97.8±1.7	98.2±0.9	96.3±2.8	96.4±2.8	96.4±2.6	96.8±1.5	97.4±1.3	97.6±0.9	98.0±1.0	97.8±1.1	97.6±1.2
3	77.9±2.6	90.1±5.0	83.8±1.5	84.1±1.8	83.7±2.0	84.2±1.3	83.2±1.7	83.1±1.1	87.2±2.2	87.1±2.3	86.8±2.4
4	90.1±3.0	94.1±4.4	93.8±1.6	93.7±1.1	93.8±1.0	93.4±2.6	94.4±2.0	94.5±2.5	95.0±3.0	95.2±2.4	95.0±2.8
5	84.4±4.3	94.5±4.8	87.1±4.3	87.4±4.4	87.3±4.5	87.6±4.9	87.6±5.5	88.0±4.7	88.8±4.6	89.8±3.2	89.6±2.9
6	92.0±1.3	96.0±2.3	92.6±0.9	92.0±1.4	92.2±1.4	93.9±1.3	93.7±2.7	93.2±3.0	93.3±1.0	93.9±0.8	94.1±1.2
7	95.6±2.0	97.7±1.4	95.5±1.1	94.9±0.9	95.6±1.0	95.7±1.3	95.6±1.9	95.0±1.3	97.1±1.6	97.5±1.0	97.3±1.1
Avg	90.8±7.4	95.5±2.8	92.4±5.1	92.3±4.9	92.3±5.1	92.6±4.9	92.6±5.2	92.7±5.3	94.0±4.5	94.3±4.4	94.1±4.4
Sig			--	--	--	--	--	--	*	*	*

“Single Trial” and “Four Trials” indicate the classifiers were trained using data from single trial and four trials, respectively. “Real: Synthetic” means the ratio between the real data and synthetic data. “Sig” represents the significances which were conducted between the “Single Trial” and different augmentation methods (* $p < 0.05$, “--”: not significant).

detailed in Section II. For a comprehensive assessment, we compared the proposed AdaWGAN with Geometric Augmentation and AC-WGAN-GP using identical data splits and training configurations.

Table II details the gesture classification accuracy for seven subjects on the SIAT dataset. The proposed AdaWGAN consistently outperformed both the non-augmented “Single Trial” condition and other augmentation methods across all subjects and data ratios. Specifically, AdaWGAN achieved statistically significant improvements ($p \leq 0.05$) across all augmentation ratios, reaching the highest overall mean accuracy of $88.2 \pm 6.3\%$, an improvement of approximately 5% over the unaugmented model. A comparison between the “Four Trials” and “Single Trial” conditions reveals that recognition accuracy declines significantly when the training data size is reduced. However, applying AdaWGAN with even a modest

augmentation ratio effectively compensates for the limited data size, substantially recovering recognition accuracy.

Table III presents the experimental results of data augmentation on the public HD-sEMG-65 dataset. Across all subjects and augmentation ratios, the proposed AdaWGAN achieved the highest mean gesture classification accuracy of $94.3 \pm 4.4\%$, significantly outperforming the “Single Trial” condition ($p \leq 0.05$). Most subjects reached their peak classification performance at an augmentation ratio of 1:1/2 or 1:1, while the remaining subjects achieved maximum improvement at a 1:2 ratio. Notably, Subject 3 achieved the greatest improvement in classification accuracy, with an increase of 9.3%, while Subjects 4 and 5 showed improvements of approximately 5.2%. Although the magnitude of improvement varied across individuals, the consistent gains demonstrate that the proposed method effectively enhances classification performance across different subjects.

TABLE IV
CLASSIFICATION PERFORMANCE UNDER DIFFERENT REAL-TO-SYNTHETIC AUGMENTATION RATIOS USING ADAWGAN

Dataset	Single Trial	Augmentation Ratio				
		1:1/2	1:1	1:2	1:4	1:8
SIAT	83.6±7.9	88.2±6.3	87.8±6.5	87.7±6.5	87.3±6.6	86.6±6.4
HD-sEMG-65	91.9±5.4	94.0±4.5	94.3±4.4	94.1±4.4	93.1±4.4	92.9±4.7

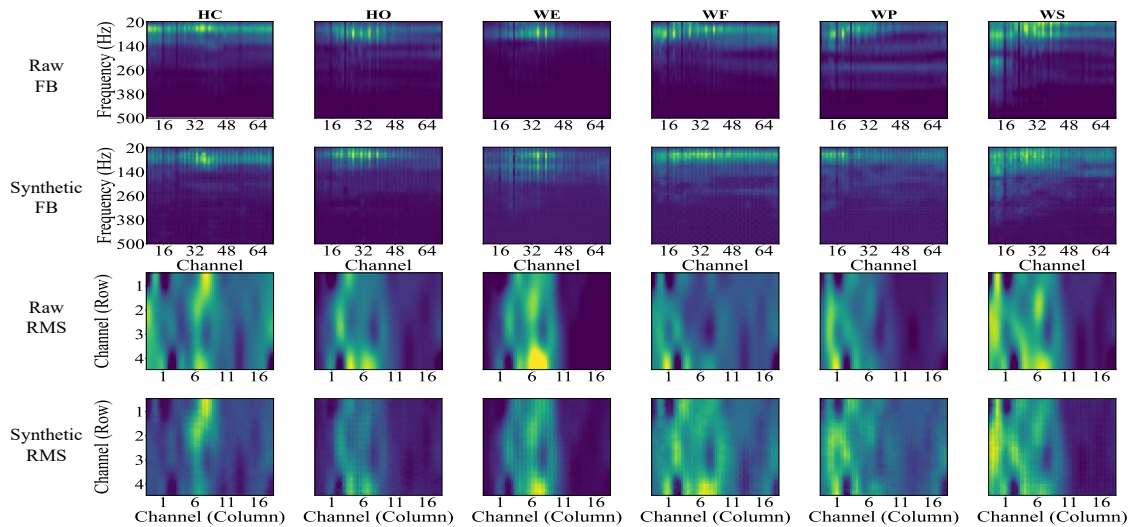


Fig. 3. A comparison of feature maps across gestures between original and AdaWGAN-Generated data. FB denotes frequency band.

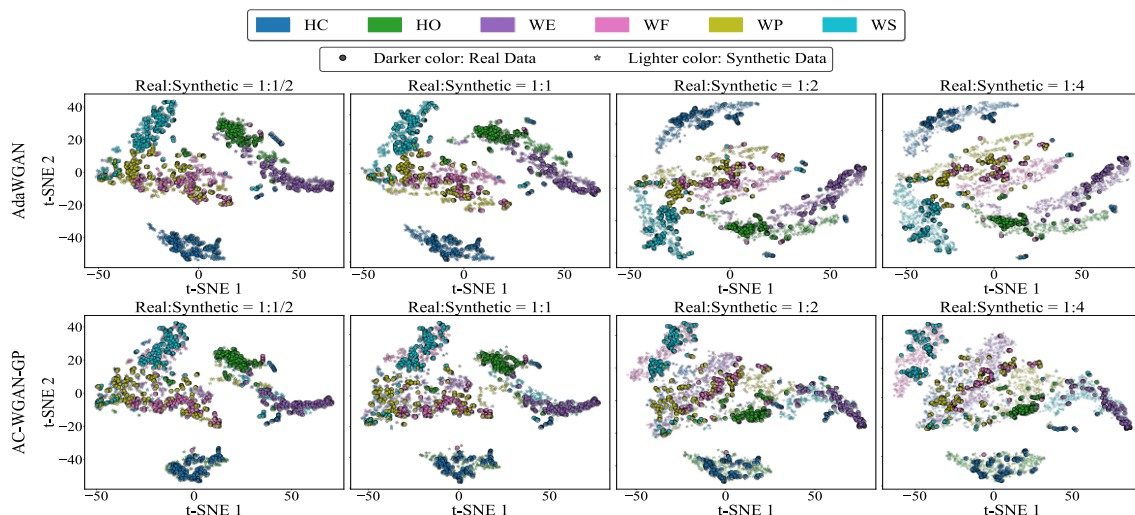


Fig. 4. 2D t-SNE visualization of the distributions of real data and synthetic data generated by AdaWGAN and AC-WGAN-GP at different mixing ratios.

To further evaluate aggressive augmentation, we extended the real-to-synthetic ratios to 1:4 and 1:8, as summarized in Table IV. The results show a non-monotonic trend on both datasets. When the ratio increased to 1:4 and 1:8, the accuracy decreased on both datasets, indicating that excessive synthetic samples may introduce redundant or biased patterns and reduce the classifier's generalization to real data.

B. Fidelity Evaluation of Generated Samples

A comprehensive evaluation of the generative quality of the proposed AdaWGAN framework was conducted using a systematic experimental approach, encompassing feature map visualization, distribution similarity, and correlation analysis. We examined the manifold distributions of real and synthetic

samples, considering the unique characteristics of each gesture class. Fig. 3 compares the feature maps of real samples with those generated by AdaWGAN for each gesture. The visualization demonstrates that the generated data preserves the activation patterns of the real data across different frequency-band (FB) maps, and maintains inter-channel response correlations in the RMS maps, particularly for gestures such as HC and WS. For the WF feature maps, slight deviations in frequency characteristics were noted in the low-frequency channels of the FB representation, with the RMS maps showing intensity variations in the lower-middle regions. Rather than being detrimental, these subtle differences serve to diversify the data distribution, effectively broadening the distribution coverage to better encompass the test set.

TABLE V
GESTURE-WISE CONSISTENCY AND AUGMENTATION EFFECT ANALYSIS

Model	Gesture	Base Acc (%)	Δ Acc (%)	Total r	RMS r	FB r	Dpre	Dpost
AdaWGAN	HC	86.2	+3.2	1.00	0.99	0.99	2.40	1.92
	HO	83.0	+1.3	0.99	0.98	0.99	1.43	1.66
	WE	87.1	+3.4	0.98	0.96	0.96	1.81	1.97
	WF	77.2	+7.6	0.95	0.72	0.86	1.77	2.14
	WP	80.9	+6.2	0.96	0.89	0.84	2.28	2.56
	WS	88.8	+3.4	0.95	0.95	0.88	3.33	3.24
AC-WGAN-GP	HC	86.2	+0.1	0.68	0.54	0.21	2.40	0.54
	HO	83.0	+0.9	0.62	0.34	0.44	1.43	0.28
	WE	87.1	+1.1	0.82	0.70	0.69	1.81	0.41
	WF	77.2	+0.2	0.83	0.20	0.36	1.77	0.24
	WP	80.9	+0.5	0.85	0.17	0.46	2.28	0.24
	WS	88.8	+0.2	0.77	0.52	0.31	3.33	0.73

Baseline Acc. denotes the accuracy before augmentation, and Δ Acc. denotes the accuracy gain after AdaWGAN-based augmentation. Total r, RMS r, and FB r denote Pearson correlations from concatenated RMS-FB, RMS, and frequency-band features, respectively. Dpre and Dpost denote intra-class feature distances before and after augmentation.

Complementing the feature map analysis, we employed t-SNE to project the high-dimensional features into a 2D space, visualizing the alignment between the original real data distribution and the synthetic data generated by both AdaWGAN and AC-WGAN-GP. The resulting distribution plots, shown in Fig. 4, use circles for real data, transparent stars for synthetic data, and different colors to represent different gestures. Each panel contains 1035 real data points, while the number of synthetic data points is determined by the corresponding real-to-synthetic ratio. With AdaWGAN augmentation, the classification boundaries become more distinct as the real-to-synthetic data ratio increases from 1:1/2 to 1:4, particularly for the HO and WE gestures. For the difficult-to-distinguish WF and WP gestures, AdaWGAN's effect is especially pronounced, with each class forming more independent and compact clusters as more synthetic data is added. In contrast, while the samples generated by AC-WGAN-GP partially preserve the structure of the original data, the absence of adaptive class weighting results in class confusion and blurred inter-class boundaries.

To quantitatively evaluate generation fidelity, we computed Pearson correlation coefficients between real and synthetic samples using averaged feature intensities within each gesture category. Correlations were calculated for RMS features, FB features, and their concatenation, denoted as RMS r, FB r, and Total r, respectively. As shown in Table V, AdaWGAN exhibits strong positive correlations with real data across all six gestures, with Total r ranging from 0.95 to 1.00. For difficult-to-distinguish gestures with lower baseline accuracies, such as WF and WP, AdaWGAN maintains high structural consistency with Total r $\geq$ 0.95. Gesture-wise analysis further shows that WF and WP achieved the largest classification gains, reaching 7.6% and 6.2%, respectively. HC, WE, and WS show moderate gains while maintaining high Total r values, indicating stable generation performance across different gesture types. In contrast, HO shows the smallest gain despite its high Total r, suggesting that highly consistent generated samples may provide limited additional diversity for this gesture. The intra-

class distance analysis further shows that AdaWGAN generally moderately expands the gesture-specific feature distributions, whereas AC-WGAN-GP sharply reduces Dpost across all gestures. This is consistent with Fig. 4, where AC-WGAN-GP samples are mixed with neighboring real classes and show blurred class boundaries, leading to limited classification improvement.

C. Biological Interpretation of the Generative Network

We employed activation-map (AM) and Integrated Gradient (IG) methods (as depicted in Section II-D) to probe and visualize the operational logic of the AdaWGAN discriminator. Fig. 5 illustrates these attribution results for a representative subject from the SIAT dataset during Hand Closure (HC) and Hand Open (HO) movements.

The left columns of Fig. 5(a) and (b) display the raw input features, while the middle columns present the activation map. These maps represent the average response heatmaps derived from the discriminator's first convolutional layer, where high response intensity indicates the specific regions on which the network focuses. As observed in the "Activation Response" maps, the discriminator exhibits spatially localized sensitivity. For instance, in the HC task (Fig. 5(a)), the activation is concentrated in the upper-central region, indicated by high-intensity red and yellow "hotspots." Conversely, the HO task (Fig. 5(b)) triggers a distinct activation pattern shifted toward the lower-left area.

The right columns in Fig. 5(a) and (b) provide Integrated Gradient (IG) visualizations, which quantify the contribution of individual pixels to the model's final prediction. Brighter areas indicate higher contributions to the discriminator's output. The high-contribution areas in the IG maps spatially coincide with the hotspots observed in the "Activation Response" maps, suggesting that the model's decision-making relies on active muscle regions rather than noise or artifacts.

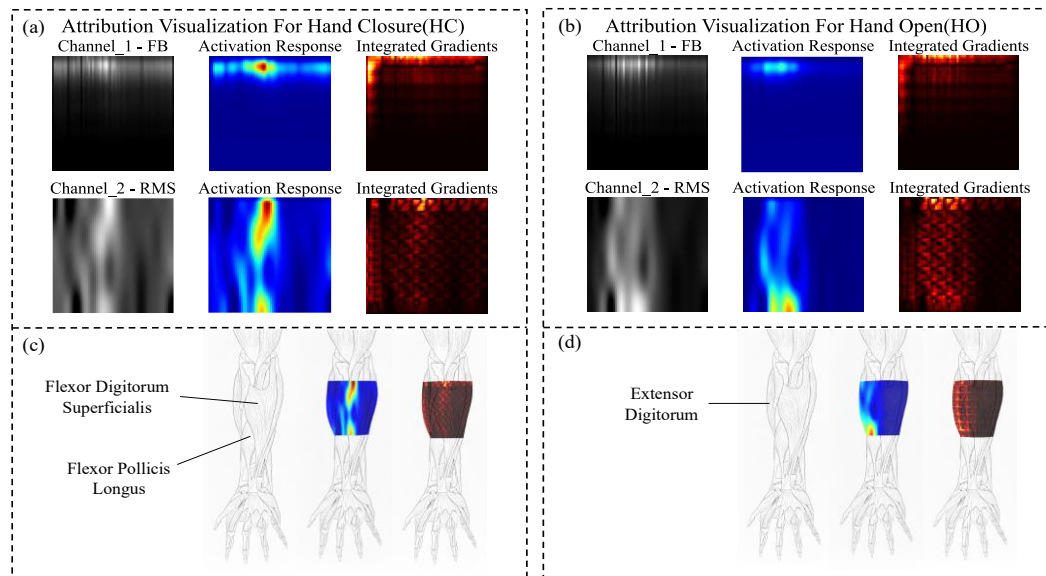


Fig. 5. Attribution analysis for each channel of motion HC (a) and HO (b); muscle groups corresponding to HC (c) and HO (d) and the layout of their visualization results on actual electrodes.

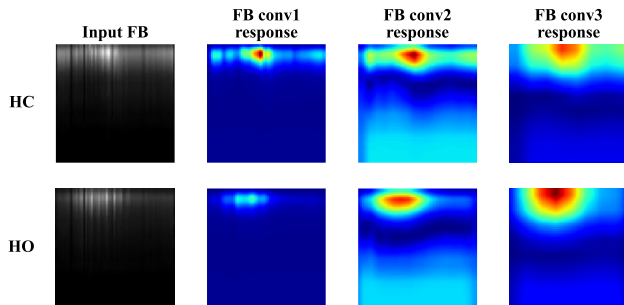


Fig. 6. Layer-wise activation responses of the frequency-band branch for HC and HO gestures. FB conv1, FB conv2 and FB conv3 denote the first, second and third convolutional layers in the frequency-band branch, respectively.

To validate the biomechanical plausibility of these learned features, we mapped the attribution results onto the anatomical structure of the forearm, as shown in Fig. 5(c) and (d). For the HC movement, the high-contribution regions align accurately with the flexor digitorum superficialis and flexor pollicis longus, the primary muscle groups responsible for finger flexion. Similarly, for the HO movement, the attribution highlights the extensor digitorum, which drives finger extension. The high degree of concordance between the discriminator's focused regions and the actual physiological activation zones demonstrates that AdaWGAN has learned biomechanically meaningful feature representations, ensuring that the generated synthetic data preserves the authentic semantic structure of human motor intent.

To examine spatial-context learning in the frequency representation, we further visualized deeper activation responses from the frequency-band branch. As shown in Fig. 6, FB Conv1 mainly captures localized frequency responses, whereas FB Conv2 and FB Conv3 layer show broader and more continuous activation regions. This indicates that the discriminator learns higher-level spatial-frequency contextual representations by combining local frequency-band responses

across deeper layers, rather than relying only on the first convolutional layer.

IV. DISCUSSION

A. Gesture Classification Performance and Augmentation Analysis

The effectiveness of generator-based data augmentation hinges on whether the synthesized data genuinely improves classifier performance without altering the underlying data distribution. To this end, we presented a systematic evaluation of AdaWGAN and other augmentation techniques in the context of single-trial HD-sEMG gesture classification. AdaWGAN consistently achieved top performance on both datasets, with mean accuracies of $88.2 \pm 6.3\%$ (SIAT) and $94.3 \pm 4.4\%$ (HD-sEMG-65). The performance gain showed a non-monotonic trend with increasing augmentation ratios. AdaWGAN achieved the best performance at moderate ratios, namely 1:1/2 on the SIAT dataset and 1:1 on the HD-sEMG-65 dataset. This aligns with existing literature [13], [25] and our findings in Table II and III. When the ratio increased aggressively to 1:4 and 1:8, the accuracy decreased on both datasets. These results indicate that moderate augmentation is more beneficial than simply increasing the number of synthetic samples. Although the generated samples preserve high fidelity to real data, they are still approximations of the true HD-sEMG distribution. When synthetic samples dominate the training set, redundant or biased generated patterns may shift the training distribution and reduce the classifier's generalization to real test data.

Notably, the HD-sEMG-65 dataset demonstrated the capacity to achieve maximal performance gains even at higher augmentation ratios. This distinct behavior stems from the inherent disparity between the two datasets [28], [29]. Specifically, HD-sEMG-65 features a higher sampling ssfrequency and longer duration for single-trial gesture

recordings. In addition, it uses a high-density 128-channel electrode setup, which is twice the channel count of the SIAT dataset. This configuration results in significantly greater information density and richer spatial resolution. This richer representation provides the generative network with more diverse samples during training, enabling the synthesis of highly informative data. Consequently, this facilitates more aggressive augmentation strategies without introducing noise or causing a distribution shift from the real data.

Consequently, moderate augmentation proved to be the most effective strategy, successfully addressing data scarcity without distorting the original distribution. Ultimately, AdaWGAN demonstrates its adaptability by delivering superior gains across both data-scarce (SIAT) and data-sufficient (HD-sEMG-65) environments. The consistent performance across varying dataset volumes confirms that the generated samples are highly realistic, effectively enhancing diversity and coverage regardless of the original dataset size. This underscores the method's robustness and wide applicability as a few-shot bio-signal augmentation tool.

B. Fidelity Evaluation of Generated Samples

We introduced a dual-branch feature representation integrating time- and frequency-domain information to provide comprehensive supervision for the generative network. A corresponding dual-stream AdaWGAN architecture was developed, enabling precise modeling of both domains to accurately capture the true sEMG data distribution [30]. Fig. 3 shows that AdaWGAN-generated samples closely matched real data in regional energy and channel-wise intensity without cross-feature interference, affirming the dual-stream design. Although minor local intensity differences were present in each feature map between the synthetic and real data, the overall spatial distribution remained highly consistent. This indicates that the generated samples faithfully replicate the fundamental characteristics of real data while simultaneously introducing controlled variability, which is a crucial factor for effective data augmentation that broadens the representational space.

Fig. 4 demonstrates that the generated samples formed natural extensions around real data clusters while maintaining sharp distribution boundaries across various augmentation ratios. These results indicate that AdaWGAN successfully captured intra-gesture distributions and enhanced diversity while preserving inter-gesture separability [31], [32], underscoring its robustness for few-shot data augmentation. However, as the ratio increases, the synthetic data distribution tends to disperse outward, rather than clustering compactly around the center of the real data. When the real-to-synthetic ratio increases to 1:2, the synthetic samples show increased dispersion and partial class overlap, particularly for gestures such as HC and HO under AC-WGAN-GP. This trend is further supported by Table V, where AC-WGAN-GP shows lower correlation scores for HC and HO, with r -scores of 0.68 and 0.62, respectively. These lower correlations indicate weaker consistency between real and generated feature distributions, which may contribute to the reduced classification

improvement of these gestures. In contrast, AdaWGAN maintained higher correlation scores for HC and HO and showed more stable cluster structures, suggesting that the adaptive weighting mechanism helps preserve gesture-specific semantic consistency during generation. These phenomena explain the decline in gesture classification accuracy, as evidenced in Table II and III. This suggests that excessive augmentation introduces new noise to the classifier, resulting in performance degradation.

The strong correlation results shown in Table V confirm that AdaWGAN captured the underlying semantics of the gestures, rather than merely memorizing training examples. For gestures with lower baseline accuracies, such as WF and WP, AdaWGAN maintains high Total r values while introducing moderate RMS/FB variation and controlled distribution expansion, resulting in larger accuracy gains. In contrast, WS has the highest baseline accuracy and largest pre-augmentation intra-class distance, suggesting that its original samples already contain rich variations and therefore gain less from augmentation. HO shows the smallest gain despite very high feature consistency, indicating that overly similar generated samples may provide limited additional diversity. In comparison, AC-WGAN-GP exhibits weaker correlations and reduced intra-class distances, indicating overlapping feature distributions in the generated samples. This effect is particularly evident for WF, WP, and WS, where the RMS/FB correlations and intra-class distances decrease markedly, resulting in little accuracy improvement. Although HO and WE show relatively more stable correlations under AC-WGAN-GP, their correlations remain much lower than those achieved by AdaWGAN, leading to smaller accuracy gains. These results indicate mode collapse and a failure to learn discriminative features, ultimately leading to ambiguous sample generation and weaker correlations [33]. These results demonstrate that the adaptive weighting mechanism is critical for maintaining class-semantic consistency and improving classification performance.

C. Biological Interpretation of the Generative Network

We used attribution visualization methods to analyze the average responses of convolutional kernels and the accumulated gradients of the model. This enables an interpretability analysis of the internal perceptual mechanisms within the network [34]. The interpretability analysis presented in Section III offers critical insights into the internal mechanisms of AdaWGAN. While deep generative models are often criticized as black boxes, the high concordance between the discriminator's attribution maps and the anatomical muscle groups (Fig. 5(c) and (d)) serves as a strong validation of our model's reliability. The discriminator bases its decisions on the activation of the flexor and extensor groups rather than background noise or channel-specific artifacts, ensuring that the adversarial training process is biologically grounded. The deeper frequency-branch visualization further supports this interpretation, indicating that AdaWGAN learns spatial-frequency context from frequency-band representations rather than relying only on isolated shallow activations. Consequently,

this imposes a strict physiological constraint on the generator. To deceive such a discriminator, the generator is compelled to synthesize samples that possess not only statistical similarity but also physiologically plausible spatiotemporal distributions corresponding to actual muscle synergies. This mechanism guarantees that the synthesized data preserves the authentic semantic structure of human motor intent and ensures its viability for downstream applications, such as dataset augmentation for gesture classification.

D. Limitations and Future Work

A primary limitation of this work is that the data used in this study were exclusively collected from able-bodied subjects, and thus the applicability and generalizability of the proposed method to target patient populations, including amputees and stroke patients, remain to be validated. To address this, we will recruit more target patients to evaluate the efficacy of our AdaWGAN-based data augmentation framework. Another limitation is that the proposed AdaWGAN framework comprises three submodules, resulting in a relatively complex architecture. Moreover, the training procedure involves multiple loss functions and their coordinated optimization, which further increases the computational burden. For practical HCI applications, improving computational efficiency is important for rapid calibration and real-time-oriented deployment. In future work, we plan to reduce the structural complexity of the model, explore lightweight and quantized implementations, and refine the training objectives, thereby lowering the computational cost and improving overall deployment efficiency.

V. CONCLUSION

This study addressed the scarcity of surface electromyography (sEMG) data caused by its laborious acquisition process by proposing a generative model to synthesize high-fidelity data for training set augmentation. By leveraging complementary time-frequency inputs, AdaWGAN enhances generation quality through its adaptive weighting mechanism. This mechanism dynamically balances the learning process, allowing the model to capture complex activation patterns with precision. This ensures that the generated data preserve the genuine physiological characteristics and distinct class specificity required for effective data augmentation. In single-trial gesture classification experiments augmented by the synthesized samples, our method significantly improved performance on both datasets, achieving average accuracies of $88.2 \pm 6.3\%$ and $94.3 \pm 4.4\%$, respectively, demonstrating a stable performance improvement. To further investigate the model's interpretability and physiological plausibility, we introduced neural network attribution visualization methods to systematically analyze the discriminator's response to key physiological features, validating the consistency between the generation process and real muscle activation patterns. Overall, this work provides an effective solution for augmenting few-shot physiological signal datasets, and offers new insights for

building interpretable generative neural network models that align with physiological mechanisms.

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